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Experiment – 1

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Subject: Full Stack - II **Subject Code:** 24CSP-337

EXPERIMENT – 1.1

1. **AIM:**
   To design and develop a dynamic post composer interface supporting multiple platforms with constraint validation.

2. **OBJECTIVE:**
   1. To understand multi-platform content handling.
   2. To implement real-time validation mechanisms.
   3. To design responsive and user-friendly UI components

3. **SOFTWARE REQUIREMENTS:**
   1. Windows/Linux/macOS
   2. Node.js (LTS Version)
   3. npm (Node Package Manager)
   4. Visual Studio Code
   5. Google Chrome / Microsoft Edge
   6. React.js
   7. Command Prompt / Terminal

4. **THEORY**

   React is an open-source JavaScript library developed by Meta (Facebook) for building fast, dynamic, and interactive user interfaces. It follows a component-based architecture, where the user interface is divided into reusable components. This approach improves code reusability, readability, and maintainability.

   In modern social media applications, users often create content for multiple platforms such as Twitter, LinkedIn, and Instagram. Each platform has different character limits and posting rules. Therefore, a Post Composer is designed to help users create content while validating platform-specific constraints before publishing. Real-time validation improves user experience by immediately informing users if the entered content exceeds the allowed limit.

   The Post Composer developed in this experiment is implemented using React Functional Components and React Hooks. The useState() hook is used to store and manage dynamic data such as the selected platform and the content entered by the user. Whenever the state changes, React automatically updates the user interface without reloading the entire webpage.

   User interactions such as typing in the textarea or selecting a platform are handled using event handling (onChange). The application also uses controlled components, where the value of form elements is managed through React state. This ensures synchronization between the user interface and the application's data.

The application dynamically calculates the number of characters entered using the length property and compares it with the predefined character limit of the selected platform. Based on this comparison, conditional rendering is used to display appropriate validation messages such as "Ready to Post" or "Character Limit Exceeded."

Overall, this experiment demonstrates the practical implementation of React concepts including components, JSX, state management, event handling, controlled components, dynamic rendering, and real-time validation, providing the foundation for building modern frontend applications.

5. IMPLEMENTATION/PROCEDURE

- Install Node.js and verify the installation using node -v and npm -v.
- Create a React project using npx create-react-app post-composer.
- Open the project in Visual Studio Code and run it using npm start.
- Design the Post Composer interface with a heading, platform dropdown, and textarea.
- Implement React State (useState) to manage the selected platform and post content.
- Add a live character counter using the length property.
- Define platform-specific character limits (Twitter, LinkedIn, and Instagram).
- Implement real-time validation to display appropriate messages based on the selected platform's character limit.
- Test the application by entering different post lengths and switching between platforms.
- Verify that the application displays the correct character count and validation message for each platform.

6. CODE

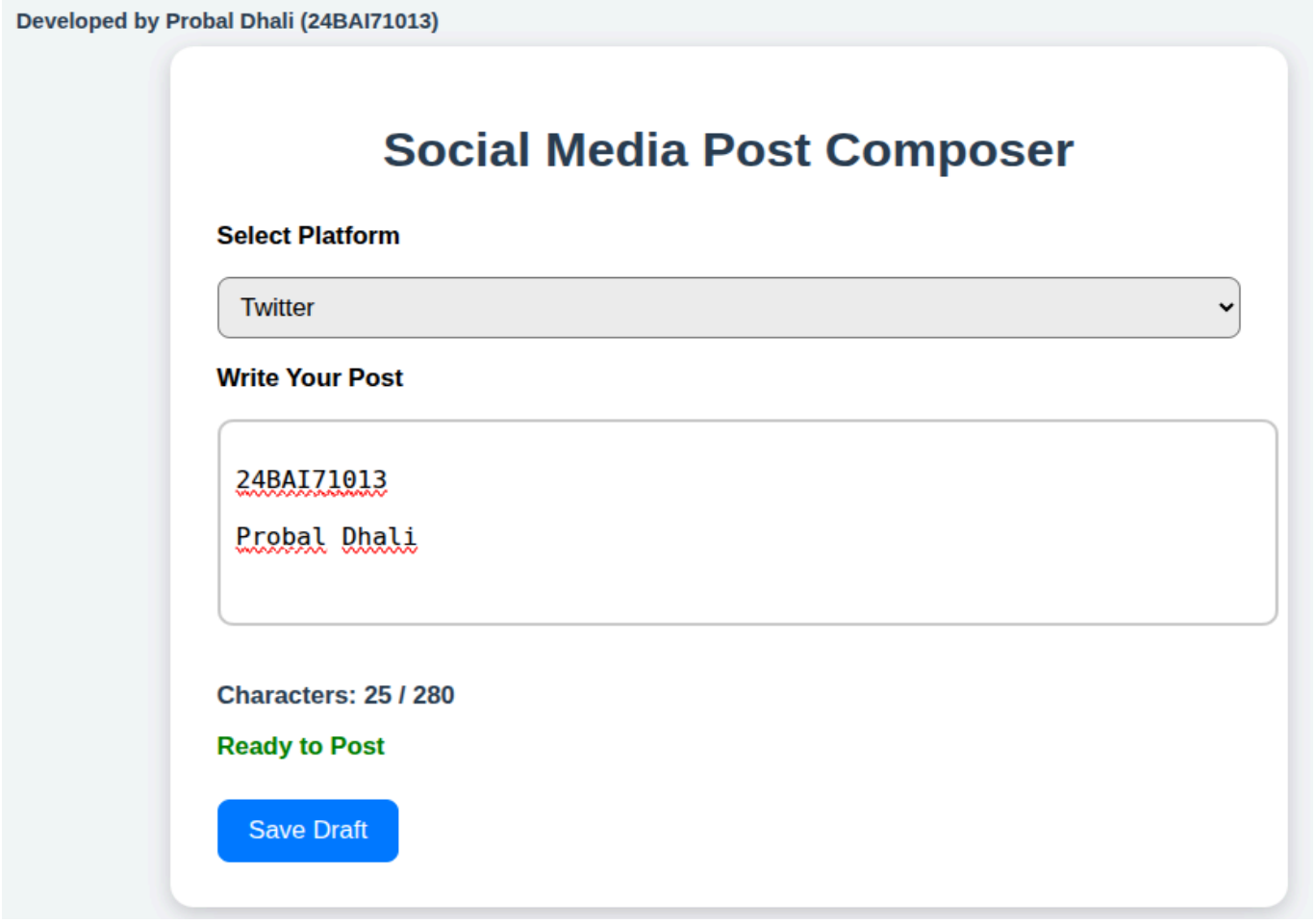
```
import { useState } from "react";

function App() {
 const [platform, setPlatform] = useState("Twitter");
 const [post, setPost] = useState("");

 const limits = {
  Twitter: 280,
  LinkedIn: 3000,
  Instagram: 2200,
  Facebook: 63206,
 };

 const limit = limits[platform];
 const isExceeded = post.length > limit;

 return (
  <div
   style={{
    backgroundColor: "#f4f6f9",
    minHeight: "100vh",
    padding: "40px",
    fontFamily: "Arial, sans-serif",
    position: "relative",
   }}
  >
   {/* Top-left corner developer credit */}
   <div
    style={{
     position: "absolute",
     top: "15px",
     left: "20px",
     fontSize: "14px",
```

```
    fontWeight: "bold",
    color: "#2c3e50",
  }}
>
  Developed by Probal Dhali (24BAI71013) Experiment 1.1.1
</div>

<div
  style={{
    width: "650px",
    margin: "auto",
    backgroundColor: "#ffffff",
    padding: "30px",
    borderRadius: "15px",
    boxShadow: "0px 4px 15px rgba(0,0,0,0.2)",
  }}
>
  <h1
    style={{
      textAlign: "center",
      color: "#2c3e50",
      marginBottom: "30px",
    }}
  >
    Social Media Post Composer
  </h1>

  <label>
    <b>Select Platform</b>
  </label>

  <br />
  <br />

  <select
    value={platform}
    onChange={(e) => setPlatform(e.target.value)}
    style={{
      width: "100%",
      padding: "10px",
      borderRadius: "8px",
      fontSize: "16px",
    }}
  >
    <option>Twitter</option>
    <option>LinkedIn</option>
    <option>Instagram</option>
    <option>Facebook</option>
  </select>

  <br />
  <br />

  <label>
    <b>Write Your Post</b>
  </label>
```

```
<br />
<br />

<textarea
  rows="6"
  placeholder="Type your post here..."
  value={post}
  onChange={(e) => setPost(e.target.value)}
  style={{
    width: "100%",
    padding: "10px",
    borderRadius: "10px",
    border: "2px solid #ccc",
    fontSize: "16px",
    resize: "none",
  }}
></textarea>

<br />
<br />

<p
  style={{
    fontWeight: "bold",
    color: isExceeded ? "red" : "#2c3e50",
  }}
>
  Characters: {post.length} / {limit}
</p>

{isExceeded ? (
  <p
    style={{
      color: "red",
      fontWeight: "bold",
    }}
  >
    Character Limit Exceeded!
  </p>
) : (
  <p
    style={{
      color: "green",
      fontWeight: "bold",
    }}
  >
    Ready to Post
  </p>
)}

<button
  style={{
    backgroundColor: "#007bff",
    color: "white",
    border: "none",
```

```
            padding: "12px 20px",
            borderRadius: "8px",
            fontSize: "16px",
            cursor: "pointer",
            marginTop: "10px",
          }}
        >
          Save Draft
        </button>
      </div>
    </div>
  );
}

export default App;
```

7. OUTPUT



- Fig 1.1: Social Media Post Composer User Interface

The figure shows the **Social Media Post Composer** developed using **React.js**. It allows users to select a social media platform, write a post, and view the character count in real time. The application validates the entered content against the selected platform's character limit and displays the message **"Ready to Post"** when the post is within the allowed limit. A **"Save Draft"** button is also provided to save the post for future editing.

8. LEARNING OUTCOME

- Understand the fundamentals of React.js and component-based development.
- Create and run a React application using Node.js and npm.
- Design a dynamic Post Composer interface using React.
- Implement state management using the useState() Hook.
- Handle user input using controlled components and event handling.
- Implement platform-specific character validation and real-time feedback.
- Develop interactive and responsive frontend applications using React.

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EXPERIMENT – 1.2

1. AIM:

To implement a draft management system that allows users to save, retrieve, and manage post drafts within the frontend, with optional simulation of backend interactions.

2. OBJECTIVE:

1. To manage draft data using frontend state
2. To implement CRUD operations for draft handling
3. To understand asynchronous workflows in UI applications
4. To optionally simulate backend communication using mock APIs

3. SOFTWARE REQUIREMENTS:

1. Windows/Linux/macOS
2. Node.js (LTS Version)
3. npm (Node Package Manager)
4. Visual Studio Code
5. Google Chrome / Microsoft Edge
6. React.js
7. Command Prompt / Terminal

4. THEORY

A Draft Management System enables users to temporarily save content before publishing it. It improves user experience by allowing posts to be edited, updated, or deleted at any time. In this experiment, React's useState() Hook is used to manage draft data, while useEffect() and Local Storage are used to persist drafts even after refreshing the browser. The application also demonstrates CRUD operations, dynamic rendering using map(), and array manipulation using filter(), providing practical knowledge of frontend state management**.**

5. IMPLEMENTATION/PROCEDURE

- Create a React State to store multiple drafts.
- Implement the Save Draft functionality using useState().
- Display all saved drafts using the map() function.
- Implement Edit Draft functionality.
- Implement Delete Draft functionality using filter().
- Store drafts in the browser using Local Storage.
- Load saved drafts automatically using useEffect().
- Test all CRUD operations.

6. CODE

```
import { useState } from "react";

function App() {
  const [platform, setPlatform] = useState("Twitter");
  const [post, setPost] = useState("");
  const [drafts, setDrafts] = useState([]);

  const limits = {
    Twitter: 280,
```

```
  LinkedIn: 3000,
  Instagram: 2200,
};

const limit = limits[platform];
const isExceeded = post.length > limit;

const saveDraft = () => {
  if (post.trim() === "") {
    alert("Please write something first!");
    return;
  }

  setDrafts([...drafts, post]);
  setPost("");
};

const editDraft = (index) => {
  setPost(drafts[index]);

  const updatedDrafts = drafts.filter((_, i) => i !== index);
  setDrafts(updatedDrafts);
};

const deleteDraft = (index) => {
  const updatedDrafts = drafts.filter((_, i) => i !== index);
  setDrafts(updatedDrafts);
};

return (
  <div
    style={{
      backgroundColor: "#f4f6f9",
      minHeight: "100vh",
      padding: "40px",
      fontFamily: "Arial",
    }}
  >
    {/* Top-left corner developer credit */}
    <div
      style={{
          position: "absolute",
        top: "15px",
        left: "20px",
        fontSize: "14px",
        fontWeight: "bold",
        color: "#2c3e50",
      }}
    >
      Developed by Probal Dhali (24BAI71013) Experiment 1.1.2
    </div>

    <div
      style={{
        backgroundColor: "white",
        width: "650px",
```

```
    margin: "auto",
    padding: "30px",
    borderRadius: "15px",
    boxShadow: "0px 4px 15px rgba(0,0,0,0.2)",
   }}
  >
   <h1
    style={{
     textAlign: "center",
     color: "#2c3e50",
     marginBottom: "30px",
    }}
   >
    Social Media Post Composer
   </h1>

   <label>
    <b>Select Platform</b>
   </label>

   <br />
   <br />

   <select
    value={platform}
    onChange={(e) => setPlatform(e.target.value)}
    style={{
     width: "100%",
     padding: "10px",
     borderRadius: "8px",
     fontSize: "16px",
    }}
   >
    <option>Twitter</option>
    <option>LinkedIn</option>
    <option>Instagram</option>
   </select>

   <br />
   <br />

   <label>
    <b>Write Your Post</b>
   </label>

   <br />
   <br />

   <textarea
    rows="6"
    placeholder="Type your post here..."
    value={post}
    onChange={(e) => setPost(e.target.value)}
    style={{
     width: "100%",
     padding: "10px",
```

```
    borderRadius: "10px",
    border: "2px solid #ccc",
    fontSize: "16px",
   }}
></textarea>

<br />
<br />

<p
  style={{
    fontWeight: "bold",
    color: isExceeded ? "red" : "#2c3e50",
  }}
>
  Characters: {post.length} / {limit}
</p>

{isExceeded ? (
  <p
    style={{
      color: "red",
      fontWeight: "bold",
    }}
  >
    Character limit exceeded!
  </p>
) : (
  <p
    style={{
      color: "green",
      fontWeight: "bold",
    }}
  >
    Ready to post
  </p>
)}

<button
  onClick={saveDraft}
  style={{
    backgroundColor: "#007bff",
    color: "white",
    border: "none",
    padding: "12px 20px",
    borderRadius: "8px",
    cursor: "pointer",
    fontSize: "16px",
    marginTop: "10px",
  }}
>
  Save Draft
</button>

<hr style={{ margin: "30px 0" }} />
```

```
        <h2>Saved Drafts</h2>

        {drafts.length === 0 ? (
          <p>No drafts available.</p>
        ) : (
          drafts.map((draft, index) => (
            <div
              key={index}
              style={{
                backgroundColor: "#f8f9fa",
                padding: "15px",
                borderRadius: "10px",
                marginBottom: "15px",
                boxShadow: "0px 2px 8px rgba(0,0,0,0.1)",
              }}
            >
              <p>{draft}</p>

              <button
                onClick={() => editDraft(index)}
                style={{
                  backgroundColor: "#28a745",
                  color: "white",
                  border: "none",
                  padding: "8px 15px",
                  borderRadius: "5px",
                  cursor: "pointer",
                  marginRight: "10px",
                }}
              >
                ✏ Edit
              </button>

              <button
                onClick={() => deleteDraft(index)}
                style={{
                  backgroundColor: "#dc3545",
                  color: "white",
                  border: "none",
                  padding: "8px 15px",
                  borderRadius: "5px",
                  cursor: "pointer",
                }}
              >
                🗑 Delete
              </button>
            </div>
          ))
        )}
      </div>
    </div>
  );
}

export default App;
```

7. OUTPUT

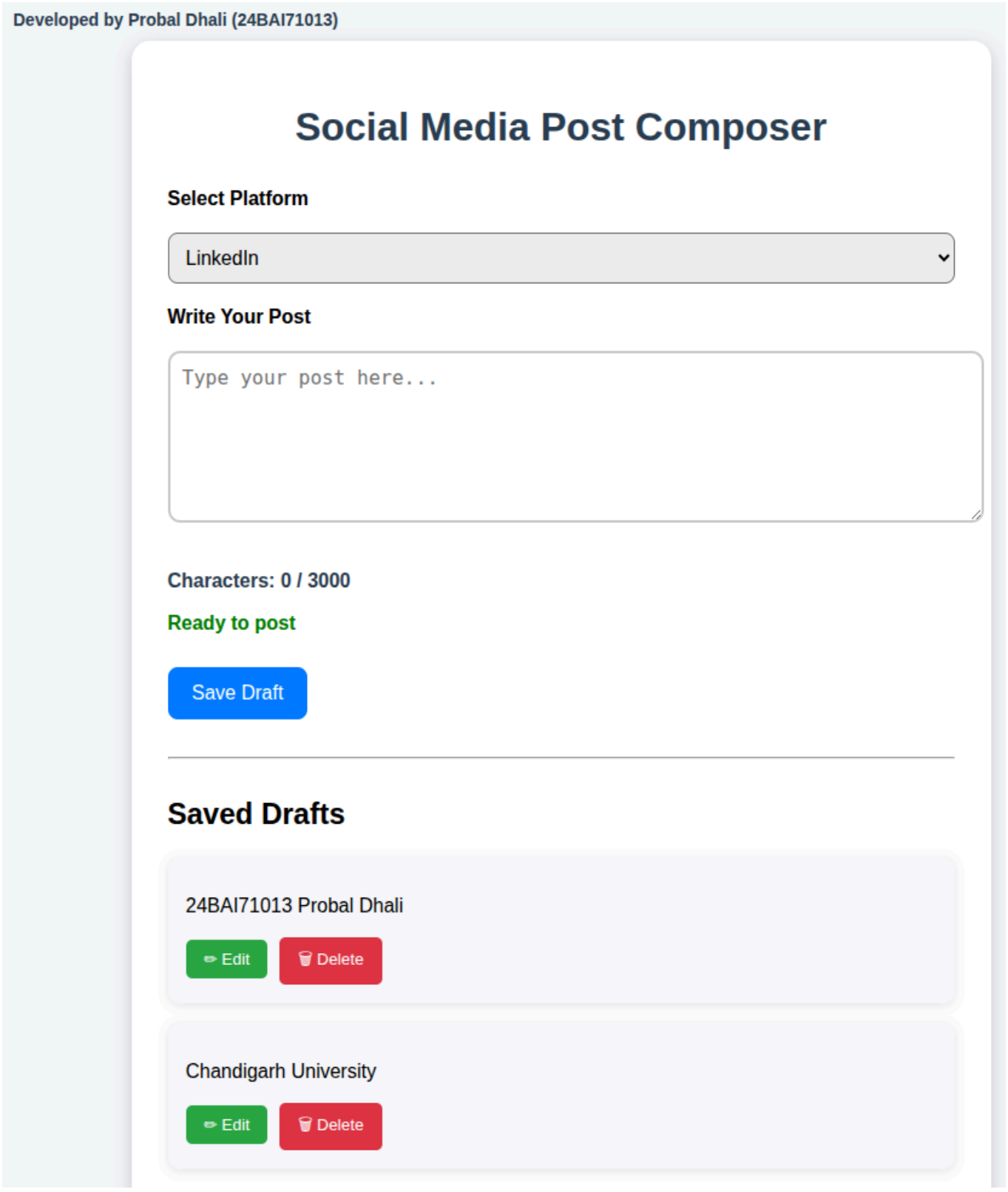

Fig 1.2: Draft Management System Webpage

The figure shows the **Draft Management System** of the Social Media Post Composer developed using **React.js**. The interface allows users to select a social media platform, compose a post, and save it as a draft. The application displays the selected platform's character limit and provides real-time validation of the entered content.

The **Saved Drafts** section lists all previously saved posts. Each draft is displayed inside a separate card with **Edit** and **Delete** buttons. The **Edit** button loads the selected draft back into the text area for modification, while the **Delete** button permanently removes the draft from the list. This interface demonstrates the implementation of **React Hooks (useState), state management, event handling, conditional rendering, and CRUD operations (Create, Read, Update, Delete)** for efficient draft management.

8. LEARNING OUTCOME

- Understand the concept of Draft Management.
- Implement CRUD operations in React.
- Manage application data using React State.
- Display dynamic data using the map() function.
- Modify arrays using the filter() function.
- Use the useEffect() Hook for lifecycle management.
- Store and retrieve data using Local Storage.
- Develop interactive frontend applications using React.